

T3 UV Lagrangian terms grouped by field content

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. The BSM-only UV expression is preferred; the full UV Lagrangian is used as a fallback when the BSM-only conversion is unavailable.

Field combinations 1 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1 D$	S_2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$		$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$
								$+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4} \lambda_{S_1}^{(1)}$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4} \lambda_{S_2}^{(1)}$

Field combinations 2 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	D	F	F	D
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1} \left(S_{2,j_1} \right)^\dagger D_{\mu_1} \left(S_{2,j_1} \right)$		$-\frac{1}{2} F_{r_1}{}^T C F_{r_1} M_F$		$+\frac{1}{2} i F_{r_1}{}^T C \gamma^{\mu_1} D_{\mu_1} \left(F_{r_1} \right)$

Field combinations 3 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	H	S_1	S_2	S_2	H
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}$	$+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^1\lambda_{12}^{(1)}$	$+H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{H2}^{(1)}$			
								$+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}$	$+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^2\lambda_{12}^{(2)}$	$+H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{H2}^{(2)}$			

Field combinations 4 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	H	S_1	F	ℓ	S_2	F	ℓ
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\overline{\lambda_{T3}} S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$			$+\overline{L_{a_1,i_1}} P_R F_{r_1} S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1}$			$+\overline{y_{2,i_1,r_1}} F_{r_1}^T C P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I}_{y_2,a_1,j_1}$		
								$+H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$			$+\overline{y_{1,i_1,r_1}} F_{r_1}^T C P_L L_{a_1,i_1} \mathcal{I}_{y_1,a_1,j_1} S_{1,j_1}^\dagger$			$+\overline{L_{a_1,i_1}} P_R F_{r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1} S_{2,j_1}^\dagger$		